

Reduced Tillage and No-Tillage

Spatial scale: vineyard

About this sustainable management strategy

Reduced tillage and no-tillage are soil management practices that minimise mechanical soil disturbance while maintaining continuous soil cover. In vineyards, these practices are most commonly implemented in the inter-rows and are often combined with permanent or semi-permanent cover crops and mulching. Reduced tillage limits the frequency, depth, and intensity of soil cultivation, while no-tillage avoids soil disturbance altogether except for essential planting or maintenance operations.

By reducing soil disturbance, these practices help preserve soil structure, enhance biological activity, and promote the accumulation of organic matter. They can improve water infiltration, reduce soil erosion, increase water retention, and support the long-term fertility and resilience of vineyard soils. The successful implementation of reduced tillage and no-tillage systems requires management practices adapted to local soil, climate, and vineyard conditions.

Key ecosystem services

- Soil health and fertility (*improved soil structure, nutrient cycling and organic matter content*)
- Erosion control (*slope stabilisation*)
- Water retention (*reduced surface runoff and moisture control*)
- Climate regulation (*local microclimate regulation, reduced temperature extremes*)
- Carbon sequestration
- Biodiversity enhancement (*above-and below-ground*)
- Pollination
- Natural pest control (*support for beneficial insects and natural enemies*)
- Landscape aesthetics (*distinct landscape character*)

Considerations for successful implementation

Transition of management practices

The transition from conventional tillage to reduced or no-tillage systems often requires adjustments to vineyard operations. Changes in soil cover management, machinery use, and weed control strategies should be planned carefully to ensure a smooth transition while maintaining vineyard productivity.

Weed and vegetation management

As soil disturbance is reduced, vegetation management becomes a central component of the system. The integration of cover crops, mulching, or other ecological approaches supports weed suppression while contributing to soil protection and biodiversity conservation.

Soil-specific suitability

The benefits of reduced tillage and no-tillage are strongly influenced by soil characteristics, climatic conditions, and topography. Local adaptation of management practices, supported by regular field observations, is essential for maintaining soil structure and enhancing long-term soil resilience.